



18-649 Lab 1: Testing Document

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Unit Testing Document

This document defines planned tests and expected results for the tabletop drive-by-wire system. Record measured results only after each test is run.

1. Test scope and traceability

Lab 1 requires testing of the blinker subsystem, one additional subsystem, and CAN redundancy using the fault-injection jumpers. This plan uses drivetrain control as the additional subsystem. It also includes self-test and recovery because they affect system safety.

- **Blinker subsystem:** left/right indication, automatic cancel, timing, synchronization and hazards.
 - **Drivetrain control:** throttle mapping, closed-loop speed control, brake priority, stuck-wheel behavior and invalid inputs.
 - **Self-test and recovery:** startup, LOCAL_FAIL, SYSTEM_FAIL, debounce, heartbeat loss and restoration.
 - **CAN redundancy:** dual transmission, duplicate suppression, degradation, isolation and recovery.
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2. Test setup and evidence

Run tests in unit-test mode where possible. Use simulated UDP state updates and CAN-frame injection or logging to create repeatable inputs. Record the software build, hardware revision, message IDs and team-defined values used for each run.

- **Oscilloscope:** measure end-to-end time from the cockpit UDP-receive GPIO edge to the output test point. Use it for motor PWM, dynamic braking, servo, blinkers and test-button timing.
- **USB-CAN adapter or CAN analyzer:** capture CAN A and CAN B separately. Record IDs, payloads, sequence counters, timestamps and bus load.
- **Section 8 setup check:** before CAN-02 through CAN-07, confirm each node's jumper labels and locations, CAN test-header pinout, status-LED meaning, termination configuration and CAN-analyzer connection point against the final Dual-CAN topology.

3. Blinker subsystem tests

Run the tests below with documented left/right thresholds. Capture front and rear blinker outputs on separate oscilloscope channels for timing and synchronization checks.

Test ID	Precondition and input	Expected result	Requirements
BL-01	Normal state. Left signal is off; wheel is not past the left threshold. Press left-turn button.	Front and rear left blinkers start within 100 ms. Right blinkers stay off.	S4; R2.4
BL-02	Left signal is active. Turn wheel past the left threshold, then return past it.	Left front and rear blinkers stop. No right blinker activates.	S4
BL-03	Normal state. Right signal is off; wheel is not past the right threshold. Press right-turn button.	Front and rear right blinkers start within 100 ms. Left blinkers stay off.	S4; R2.4
BL-04	Right signal is active. Turn wheel past the right threshold, then return past	Right front and rear blinkers stop. No left blinker activates.	S4

Test ID	Precondition and input	Expected result	Requirements
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BL-05	Left signal is active. Press right-turn button.	Right signal becomes active; left signal stops.	S4
BL-06	Normal state. Capture front and rear outputs for one active turn signal.	Blink period is 0.9–1.1 Hz with equal on/off time. Same-side front/rear edges differ by no more than 1 ms.	R2.4
BL-07	Trigger SYSTEM_FAIL from a valid normal state.	All four blinkers flash together at 2 Hz $\pm 10\%$, with equal on/off time. Normal turn state is overridden.	S4; R3
BL-08	Send an out-of-range steering, throttle, brake or turn-state value in unit-test mode.	System enters SYSTEM_FAIL and hazards activate.	R3

4. Drivetrain-control tests

Document the throttle-to-speed mapping, maximum speed, encoder scale, PWM limit and current thresholds before running these tests.

Test ID	Precondition and input	Expected result	Requirements
DR-01	Brake released. Send a low, mid and full throttle command.	Target speed increases monotonically with throttle. Full throttle does not exceed the documented maximum speed.	S1; Rear Zone

Test ID	Precondition and input	Expected result	Requirements
DR-02	Brake released. Command a steady target speed; record both encoder speeds and PWM.	Average encoder speed settles near the target and the controller adjusts PWM to hold it.	S1; Rear Zone
DR-03	Throttle is nonzero. Send a brake command.	PWM is disabled and dynamic braking is applied within 2 ms of cockpit UDP reception.	S2; R2.2
DR-04	Brake remains active while throttle changes.	Brake remains active; throttle cannot re-enable motor drive.	S2
DR-05	Command nonzero speed while preventing encoder transitions for at least 200 ms.	Controller increases PWM only up to the documented maximum. Record current and temperature limits during this test.	Rear Zone
DR-06	Send an invalid throttle, brake or speed-control value.	System enters SYSTEM_FAIL: PWM is disabled and dynamic braking is applied.	R3

5. Self-test and system-state tests

Use the System state chart terms INIT, NORMAL, LOCAL_FAIL and SYSTEM_FAIL. Run each test once per zone unless a test explicitly names one zone.

Test ID	Precondition and input	Expected result	Requirements
ST-01	System starts	Every zone begins in INIT with safe	R3

Test ID	Precondition and input	Expected result	Requirements
	powered off, then powers on.	outputs. It reaches NORMAL only after required peer heartbeats and valid STM32 control data are received.	
ST-02	Normal state. Single-press the cockpit, steering or drivetrain test input.	Pressed zone reaches LOCAL_FAIL within 10 ms, stops heartbeats on both buses and sets safe outputs.	R2.5; R3; R4.5
ST-03	One zone is in LOCAL_FAIL; monitor both other zones.	Other zones enter SYSTEM_FAIL within 100 ms of the press. Hazards flash and drivetrain brakes.	R2.5; R3
ST-04	Normal state. Generate two test-input presses less than 50 ms apart.	Treat the presses as one press; do not restore a failed zone.	R2.5
ST-05	A zone is in LOCAL_FAIL. Send two presses within 500 ms.	Zone resumes heartbeats. System returns to NORMAL within 500 ms after heartbeats from all zones are received again.	R3
ST-06	Cockpit receives no UDP state update for 100 ms.	Cockpit stops its heartbeat until UDP updates resume. Other zones then follow the normal missed-heartbeat failure path.	Cockpit Zone; R4.5

6. CAN-redundancy tests

Run CAN-02 through CAN-07 using the final Section 8 topology. Confirm the installed isolation jumpers, LED meanings and analyzer connections. These tests must be repeated for all applicable nodes and bus combinations.

Test ID	Fault action	Expected result	Requirements
CAN-01	Capture normal traffic on CAN A and CAN B.	Every catalog message appears on both buses with the same sequence counter. Receiver acts on the first valid copy and ignores the duplicate.	R4.4; R4.7
CAN-02	Isolate one node from CAN A only using its Section 8 jumper.	All functions continue with R2 timing. Affected CAN A connection is marked degraded by LED and heartbeat; no hazards or braking.	R4.7; R4.8(a)
CAN-03	Isolate one node from CAN B only.	Same expected result as CAN-02, for CAN B.	R4.7; R4.8(b)
CAN-04	Isolate two different nodes from opposite buses.	All three zones remain functional. Each affected connection shows degraded status.	R4.8(c)
CAN-05	Isolate one node from both buses.	Other zones enter SYSTEM_FAIL within 100 ms. Isolated node enters its local safe state.	R2.5; R4.8(d)
CAN-06	Disconnect every node from the same bus.	System continues normally on the remaining bus. Affected bus connections show degraded status.	R4.7; R4.8(e)
CAN-07	Restore any	Degraded indication clears within 100	R4.7; R4.8(f)

Test ID	Fault action	Expected result	Requirements
	removed jumper(s).	ms of traffic returning. Normal operation resumes within 500 ms without reset or power cycle.	
CAN-08	Cause or simulate CAN controller bus-off, then restore the bus.	Controller recovers automatically within 100 ms of bus restoration.	R4.8

7. Test-result record

Create one record for each executed test. Link the raw oscilloscope capture, CAN trace, log file or photo that supports the result.

Test ID	Run date	Build / hardware version	Evidence location	Pass / fail	Notes or measured result
<i>Add one row per executed test.</i>					

8. Test completion rules

- A test passes only when every expected result is observed and its timing requirement is met.
- If a test fails, record the observed behavior, measured timing, build/hardware version and follow-up issue before rerunning it.
- Update the traceability section in the Requirements Document with the final test ID and evidence link after verification.